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Review and Intercomparison of Machine Learning Applications for Short-term Flood Forecasting

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Abstract
Among natural hazards, floods pose the greatest threat to lives and livelihoods. To reduce flood impacts, short-term flood forecasting can contribute to early warnings that provide communities with time to react. This manuscript explores how machine learning (ML) can support short-term flood forecasting. Using two methods [strengths, weaknesses, opportunities, and threats (SWOT) and comparative performance analysis] for different forecast lead times (1–6, 6–12, 12–24, and 24–48 h), we evaluate the performance of machine learning models in 94 journal papers from 2001 to 2023. SWOT reveals that the best short-term flood forecasting was produced by hybrid, random forest (RF), long short-term memory (LSTM), artificial neural network (ANN), and adaptive neuro-fuzzy inference system (ANFIS) approaches. The comparative performance analysis, meanwhile, favors convolutional neural network, ANFIS, multilayer perceptron, k-nearest neighbors algorithm (KNN), hybrid, LSTM, ANN, and support vector machine (SVM) at 1–6 h; hybrid, ANFIS, ANN, and LSTM at 6–12 h; SVM, hybrid, and RF at 12–24 h; and hybrid and RF at 24–48 h. In general, hybrid approaches consistently perform well across all lead times. Trends such as hybridization, model selection, input data selection, and decomposition seem to improve the accuracy of models. Furthermore, effective stand-alone ML models such as ANN, SVM, RF, genetic algorithm, KNN, and LSTM, provide better outcomes through hybridization with other ML models. By including different machine learning models and parameters such as environmental, socio-economical, and climatic parameters, the hybrid system can produce more accurate flood forecasting, making it more effective for early warning operational purposes.

Keywords Deep learning · Flood prediction models · Data-driven forecasting · Machine learning comparison · Climate change Adaptations

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1 Introduction

A United Nations report, published in recognition of the International Day for Disaster Risk Reduction in 2020, demonstrates that climate-related hazards such as extreme precipitation and floods dominated other hazards during the twenty-first century (CRED and UNDRR 2020). As the climate warms, these events are expected to occur in greater frequency and magnitude. Thus, accurate and timely flood forecasting (FF) is important for achieving disaster resilience. For example, short-term FF (from hours to days up to an event; the focus of this manuscript) can be integrated into early warning systems. Long-term FF (from weeks to months up to an event), meanwhile, can inform policy. For example, Rajab et al. (2023) use machine learning approaches to find rainfall trends and to develop policy scenarios that reduce impacts.

For both short- and long-term forecasting, modeling approaches can be deterministic (i.e., physical), stochastic, machine learning (ML)-based, or a combination of these methods. The former builds a simplified catchment system and uses mathematical equations to describe internal hydrological processes [see, e.g., WAterSHed systems of 1D river network, 2D overland regime, and 3D subsurface media (WASH123D) from Yeh et al. (1998); Hydrologic Engineering Center-Hydrologic Modeling System (HEC-HMS) from Feldman (2000); Storm Water Management Model (SWMM) from Rossman (2010); and Soil and Water Assessment Tool (SWAT) from Arnold et al. (1998)]. Physically based models have been successfully applied to forecast overland flow (Albano et al. 2020; Gautam et al. 2019; Scarpino et al. 2018), rainfall–runoff (Hamdan et al. 2021), and a combination of overland and channel flow (Soomro et al. 2022). Some disadvantages of physically based flood models include the need for detailed information about the catchment (Mahmoodi et al. 2024). Another issue is the high model complexity (Kumar et al. 2023), which makes it challenging to generalize the flood model to other regions.

The stochastic model is a mathematical tool that incorporates randomness or uncertainty into its structure (Albano et al. 2018; Romanowicz and Osuch 2015). For floods, this approach more closely approximates reality than physically based models (Quilty et al. 2022) and can provide a range of outcomes or scenarios (Maqsood and Huang 2013). If combined with historical data (Quilty and Adamowski 2020), stochastic models can even identify underlying patterns that provide clues to future states. Some common stochastic algorithms include auto-regressive moving average (ARMA; e.g., Mohammadi et al. 2006), auto-regressive integrated moving average (ARIMA; e.g., Yan et al. 2022), generalized linear model (GLM; e.g., Al-Juboori 2022), and Monte Carlo simulation (MCS; e.g., Ullah and Akbar 2021). Stochastic models struggle, however, to represent extreme flood events, which are often happened due to heavy rainfall (e.g., Alimohammadi and Behrouz 2021). These events are devastating and pose significant risks to lives, communities, and infrastructure. Furthermore, because stochastic models are focused on specific variables and relationships, they can be less flexible in terms of new scenarios, such as evolving the behavior of floods in future climate, unlike alternative approaches.

ML models aim to mimic human intelligence by automatically extracting relevant insights from data (Karim et al. 2023). In recent years, they have shown great potential in hydrology. For example, combining simple artificial neural networks (ANNs) that is, interconnected nodes performing simple mathematical operations on an input with nonlinear historical data, flood forecasting has been possible [see examples in Mosavi et al. (2018)].

Limitations to ANNs [e.g., when doing forecasting tasks; Zhou et al. (2022)] have led to a growing interest in more complex deep learning (DL) models. These leverage many layers of interconnected nodes to learn underlying patterns in data. Advantages of DL include the provision of rapid results [see, e.g., Notarangelo et al. (2021) and Zhou et al. (2023)] and high performance, sometimes outperforming physically or stochastically based flood forecasting in terms of accuracy (Herath et al. 2021). A downside of DL, however, is a lack of interpretability in how the model functions and decisions are made. This has led to the emergence of explainable AI (XAI) methods [see, e.g., Mamalakis et al. (2020)]. An introduction to common ML and DL models in short-term FF is provided (see S1. Overview of Machine Learning and Deep Learning Methods in Flood Forecasting in the Supplementary Material).

Another approach is integrating a priori knowledge about physical and environmental processes into ML models. This can yield more physically reliable models with a greater level of interpretability. For instance, Chadalawada et al. (2020) enhanced a machine learning model with hydrological knowledge to model rainfall–runoff. Some experts prefer an approach called hybridization; combining two or more methods to leverage their strengths. Traditionally, this involved integrating different types of models, for example: physical and stochastic (Fang 2015), conceptual and stochastic (Sharma et al. 2024), or physical and ensemble learning approaches (Saber et al. 2023), to overcome the limitations of single models and improve the forecasting accuracy. Here, we look at the potential of hybridizing ML models with first principle–based models. For example, Ghaseminejad and Uddameri (2020) combined an ANN with a groundwater flow equation to model groundwater flow systems. In Akinsoji et al. (2024), issues with ANN-based flood forecasting for certain lead times were countered through coupling with a physical distributed hydrological model. In Farfán et al. (2020), it was shown that combining physical and neural network (NN)-driven models enhanced their individual capacities for flow forecasting. However, the hybridization of physically based models with ML models is computationally expensive. Therefore, this study primarily focuses on hybridizing ML models with other ML models and evaluates different stand-alone and hybrid ML models for short-term FF in different basins and at different lead times to identify the most suitable ML models based on application, results, and accuracy. In doing so, the results of this study can be helpful for those who are involved in using ML models for FF, as well as aid in the development of early warning systems to mitigate flood risks.

2 Material and Methods

The foundation of this study is a literature review of research on short-term FF using machine learning. First, the Scopus, WoS, and IEEE databases were searched for articles containing the words “flood” and predict* OR forecast* OR nowcast* and “artificial intelligence” OR “machine learning” OR “deep learning” OR “neural network” in the title, abstract, and keywords as shown in Fig. 1. The search resulted in 539 articles including duplicates and 357 articles after removal of duplicates. From these, 94 journal papers published between 2001 to 2023 were selected for in-depth qualitative and quantitative evaluation based on forecast lead times. Each of these journal papers focuses on spatial and temporal short-term FF and at various lead times using stand-alone or hybrid ML models. Table S1 (in the Supplementary Material) lists the stand-alone and hybrid machine learning models and the study site to

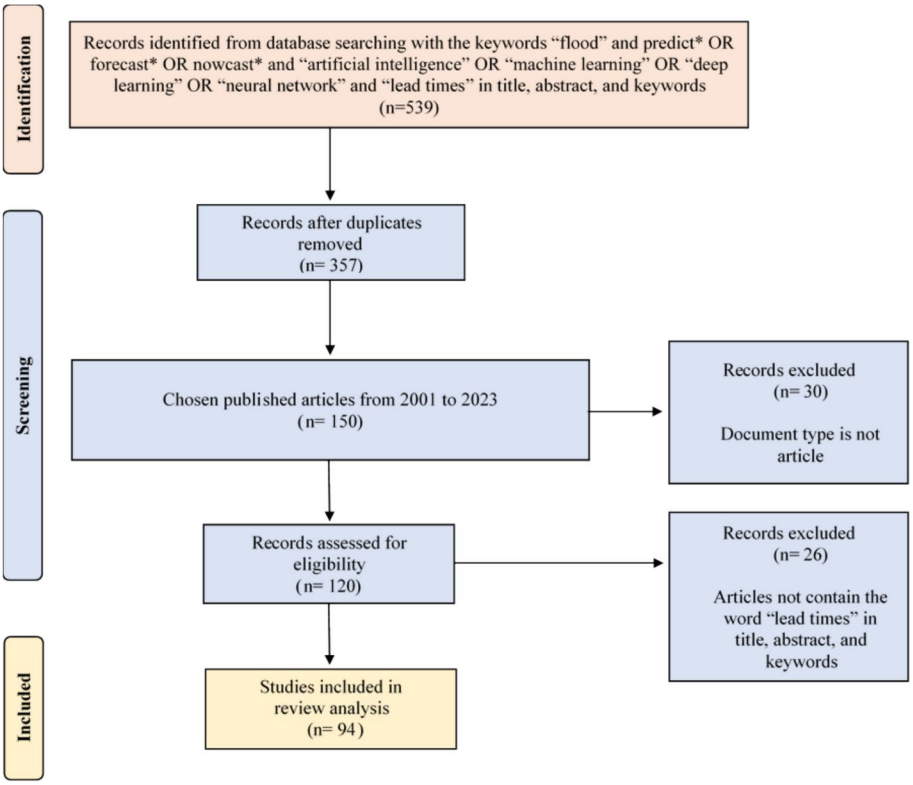


Fig. 1 PRISMA flow diagram for information search in different databases

which they are applied. Using two evaluation analyses: strengths, weaknesses, opportunities, and threats (SWOT) and comparative performance analysis (CPA), the use of ML in short-term FF across various lead times is assessed. At the end, this manuscript will provide a comprehensive performance evaluation of ML methods based on two analyses.

SWOT is a simple, transparent, and adaptable decision-making tool that is very helpful in sustainable planning and development (Morshed et al. 2023). Moreover, it has been used in many studies addressing related topics such as coastal flood management, water resources management, flash floods, stormwater, construction of dams, and early warning system (Andika and Partarini 2023; Azarnivand and Malekian 2016; Ghorbani and Hamidifar 2023; Mguni et al. 2016; Miyamoto et al. 2014). In recent decades, there has been a growing interest in applying SWOT analysis to identify and solve water managements problems (Nagara et al. 2015). However, the utilization of SWOT analysis alone is insufficient for qualitative evaluation (Oreski 2023). Görener et al. (2012) recognized this problem and made efforts to address it by integrating other methods, such as Analytic Hierarchy Process (AHP) with SWOT, or by introducing additional criteria to enhance the usability of SWOT analysis. Some studies have also introduced measures and criteria to make SWOT analysis more effective in qualitative assessment (Hamitouche and Molina 2022; Petousi et al. 2017; Sawangnate et al. 2022). These criteria provide initial information and help in conducting SWOT analysis after certain assessment. In this analysis, the predictive capability and other

performance criteria of ML methods in short-term FF across various lead times are evaluated based upon the following six criteria: 1) Simplicity: based on the number of parameters used in a model's architecture, a simple model has less parameters making it easier to implement and train. 2) Reliability: refers to method consistency and the level of uncertainty in prediction. 3) Accuracy: measure of precision between the observed and predicted data. 4) Computational time performance: effective time it takes an algorithm given a specific dataset. 5) Transparency: refers to the clarity and understandability of the model, including its decision-making processes. 6) Manageability: represents the level of complexity involved in building the model and prior expertise required to implement the model. These criteria are evaluated for the 94 studies referenced in Table S1 (in the Supplementary Material). After this evaluation, the results for each criterion are presented using qualitative rankings of high, medium, and low, as outlined by the aforementioned studies and Hamitouche and Molina (2022). While gaining some insights into overall model performance, these criteria are not conclusive in deciding which model(s) to best use for a given application. Therefore, we consider another assessment where the main strengths, weaknesses, opportunities, and threats (SWOT) of each ML method will be evaluated. These aspects can help to provide a comprehensive understanding of internal and external factors that could influence the success or failure of each model in forecasting (Chin et al. 2022; Hajizadeh 2019). Finally, the overall performance of each method was made considering the multi-criteria and SWOT aspects, aiding in the selection of the appropriate model for short-term FF. These criteria have different weights and importance in the evaluations for ML methods. In flood forecasting, the accuracy, reliability, and computation time performance are likely more important or weighted criterion than others (Kumar et al. 2023). Therefore, this study classified the performance of each ML method considering weight or importance of criterion and SWOT aspects into categories of excellent, good, average, and weak. This in-depth analysis aims to provide end users with general knowledge that can assist them when selecting ML methods for short-term FF.

In a next step, a comparative performance analysis, focused on accuracy, is conducted for the models listed in Table S1 (in the Supplementary Material). This is estimated through the normalized root mean square error (NRMSE), Nash–Sutcliffe model efficiency coefficient (NSE), and the coefficient of determination (R^2). The performance of each model is categorized as excellent, good, average, and weak based on the outcomes of the aforementioned analysis. The NRMSE (Eq. 1) is determined by dividing the difference between the maximum and minimum observed data with the root mean square error. Because root mean square error (typically provided) values can have different units (Ganthavee et al. 2024; Park et al. 2020), we use the NRMSE, given as percentages, to facilitate comparison. Values that are small or equal to 0 indicate a well-performing model (Khan and Noor 2019). The NSE and R^2, both of which are non-dimensional metrics, are calculated as shown in Eqs. 2 and 3, respectively. The NSE assesses a model's ability to predict the observed data. Values range from -∞ to 1, with higher values indicating a better capability of the model (Hussain et al. 2017). The R^2 value indicates the level of similarity between the actual and predicted values. Higher values (closer to 1) suggest a greater resemblance, whereas lower values (approaching −1) reflect greater divergence (Khan et al. 2023).

$$NRMSE = \frac{\sqrt{\frac{\sum_{i=1}^{N}(Xi,pred - Xi,obs)^2}{N}}}{Xi,obs_{max} - Xi,obs_{min}} \quad (1)$$

$$NSE = 1 - \frac{\sum_{i=1}^{N}(Xi,pred - Xi,obs)^2}{\sum_{i=1}^{N}(Xi,obs - \overline{Xobs})^2} \quad (2)$$

$$R^2 = \frac{\sum_{i=1}^{N}(Xi,pred - \overline{Xpred})(Xi,obs - \overline{Xobs})}{\sqrt{\sum_{i=1}^{N}(Xi,pred - \overline{Xpred})^2}.\sqrt{\sum_{i=1}^{N}(Xi,obs - \overline{Xobs})^2}} \quad (3)$$

In Eqs. 1–3, N is the number of samples in the dataset, *Xi,pred* and *Xi,obs* are the predicted and observed stages, while the $\overline{Xpred}$ and $\overline{Xobs}$ are the mean values of predictions and observations, respectively. The results of different ML models for short-term FF with lead times through two analyses are provided in Sect. 3.

3 Results

This section provides the outcomes of our literature review and comparison of FF methods. Figure 2 shows the most common stand-alone ML models for short-term FF applied to different basins (by country) and at different lead times (see also Table S1 in the Supplementary Material). Figure 2 (left) illustrates the growing use of ML methods in flood publications over the years from 2001 to 2023. This trend can be ascribed to advances in computational power, data availability, and the increased need to prepare for flood hazards under a changing climate. The figure also shows the emergence of new approaches like convolutional neural networks (CNN) and long short-term memory (LSTM) but ANN [including multilayer perception (MLP), back propagation neural network, and radial basis function neural network] has maintained its popularity in forecasting or as a reference for comparison. In recent years, the use of LSTM, random forest (RF), and CNN has increased due to their predictive capability features. In Fig. 2 (right), the number of studies using different ML methods is shown based on lead times. Most applications target shorter forecast horizons, which might be because longer lead times have lower model accuracy. For shorter lead times, the model can easily capture the relationships between data and make predictions

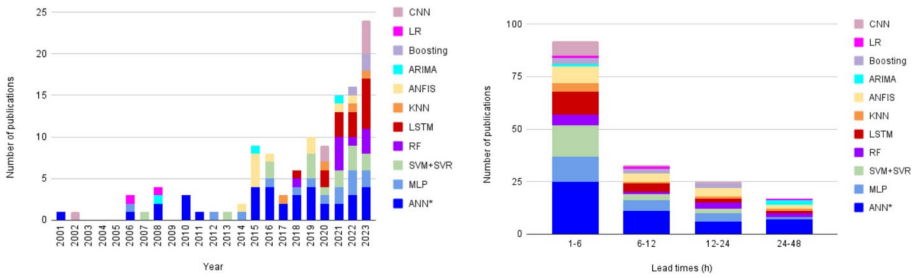


Fig. 2 Most common ML models used in short-term FF (*ANN including BPNN, MLP in addition due to its frequent use). Left: by year; right: by lead times

efficiently. However, for longer lead times, the model becomes more complex and slower due to the increased dimensionality resulting from longer time series. Therefore, model accuracy is typically higher for shorter lead times compared to longer ones. In addition, this study shows the countries that are most often targeted for case studies and in which short and longer lead times are attempted (see Figs. S1 and S2 in the Supplementary Material). The noteworthy gap in the African continent could be ascribed to gaps in observational data.

In the following, the two evaluation analyses described in Sect. 2 are used for the qualitative and quantitative performance assessment of various ML models to determine the most suitable approach for short-term FF in each lead times (see Table S1 in the Supplementary Material for an overview of all publications considered). First, the SWOT analysis is used for the qualitative performance assessment of ML models in forecasting. Second, the three statistical parameters introduced in Eqs. (1)–(3) are used for the quantitative performance evaluation.

In the SWOT analysis, six criteria are used to evaluate the predictive capabilities and operational behavior of each model. Each criterion is evaluated as high, medium, and low, based on the discussion and concluding remarks presented in the studies (see Table S1 in the Supplementary Material) that address some or all of the criteria. Additionally, insights from Hamitouche and Molina (2022) were also incorporated and summarized in Fig. 3. For simplicity, adaptive neuro-fuzzy inference system (ANFIS) and Boosting showed a high simplicity in model development. Linear regression (LR) and RF have medium simplicity. Other numerous ML models, including ANN, LSTM, CNN, support vector machine (SVM), MLP, k-nearest neighbors algorithm (KNN), ARIMA, and hybrid mostly showed low simplicity in modeling processes. For reliability, ANFIS, LSTM, MLP, ANN, and hybrid approaches showed high reliability in short-term prediction. Other approaches, such as LR, RF, SVM, and KNN demonstrated a medium reliability; whereas CNN, Boosting,

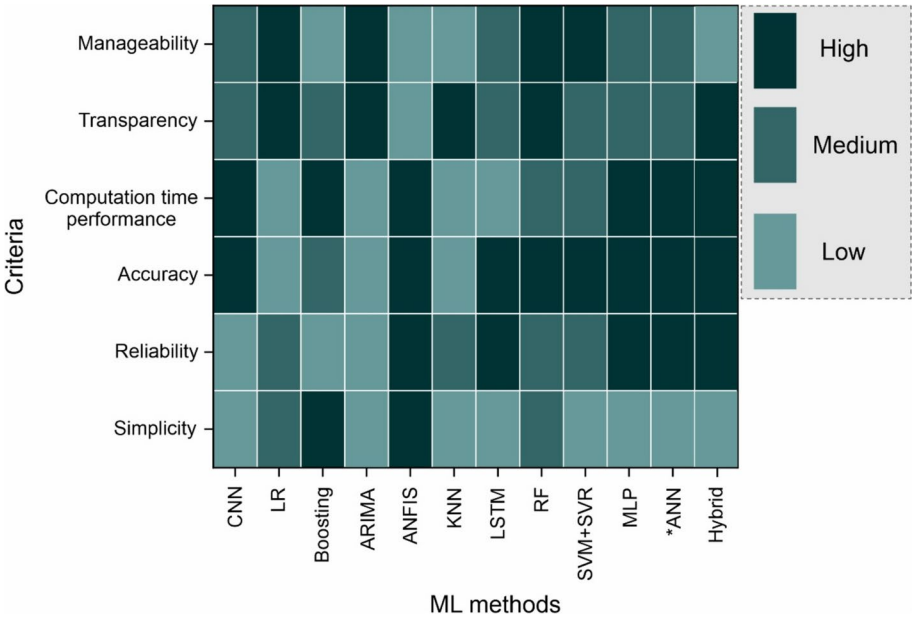


Fig. 3 Evaluation of predictive capability and working behavior of ML approaches through six criteria

and ARIMA exhibited low reliability in modeling capabilities. These models with low reliability in prediction suffered from overfitting and poor handling of nonlinearities in data. In terms of accuracy, most ML models such as hybrid, RF, ANN, MLP, SVM, LSTM, ANFIS, and CNN provided high accuracy in short-term forecasting. Boosting showed comparatively medium accuracy in predictions, while LR, ARIMA, and KNN had lower accuracy. For computation time performance, Boosting, CNN, ANFIS, MLP, ANN, and hybrid approaches showed high computation time performance due to their model structure. The remaining ML models including RF and SVM indicated medium computation time performance, whereas LR, ARIMA, KNN, and LSTM showed low performance due to their complex structure and handling of nonlinearity in data. Regarding transparency, LR, ARIMA, KNN, RF, and hybrid approaches seem best suited because they used a combination of interpretability techniques. On the other hand, CNN, Boosting, LSTM, SVM, MLP, and ANN expressed medium transparency in their forecasting capabilities due to their complexity. Finally, ANFIS demonstrated a low transparency. In terms of manageability, LR, ARIMA, RF, and SVM approaches appear to be the most manageable due to their comparatively simple structure and ease of interpretation, making them more straightforward to implement. Other ML models such as CNN, LSTM, MLP, and ANN showed medium manageability. Finally, Boosting, ANFIS, hybrid, and KNN showed low manageability due to their black-box structure and required high level of expertise for their implementation.

While the evaluation of the six criteria above provides a deeper understanding, challenges remain due to some similarities between methods (Hamitouche and Molina 2022). Therefore, a SWOT analysis is conducted for each ML method (see also Table 1). Looking at the first aspect (strength), each ML model exhibits varying strengths in forecasting. For instance, CNN, ANN, and MLP indicated strength in dealing with nonlinear input datasets and noisy environments (Kabir et al. 2020; Widiasari and Nugroho 2017). Other models, namely Boosting, SVM, and RF, exhibited a high prediction accuracy. Moreover, LR is reliable for simple classification tasks with less parameters. LSTM performed well in identifying long-term memory patterns of runoff (Kratzert et al. 2018), whereas ARIMA presented its strength in dealing with stationary and non-stationary time series data. Wang et al. (2018) developed a hybrid model including ARIMA, which showed that ARIMA could effectively predict streamflow even when dealing with stationary and non-stationary data. ANFIS showed its strength in managing model uncertainty, and the hybrid approaches provided accurate predictions. KNN showed strength in solving single and multi-objective problems. In flood modelling, KNN demonstrated its strengths in utilizing spatial proximity to classify areas as being flood prone. Moreover, KNN can effectively deal with categorization issues including several classes (Gauhar et al. 2021; Hadi et al. 2024). Regarding weaknesses, CNN needs large and high dimensional input data in simulations (Panahi et al. 2020), while RF suffers from instability issues. LR proved to be unsuitable for nonlinear relationships and the hybrid models required specialized expertise in forecasting. Moreover, the hybrid models are developed through the integration of two or more models, which require a specialist for implementation (Akbarian et al. 2022). Boosting and ANFIS face difficulties in output interpretability. Rajab (2019) mentioned the interpretability issues in fuzzy modelling during structure learning and optimization phases. ARIMA does not perform well with limited historical data during model simulation and KNN encounters difficulties in capturing the complex patterns and features. Other models like ANN, MLP, and SVM are unable to deal with complex and dynamic noisy rainfall–runoff relationships, whereas LSTM is

Table 1 Evaluation of ML in FF based on SWOT aspects

ML	Strengths	Weaknesses	Opportunities	Threats
CNN	Handling of high-dimensional and noisy inputs	High-dimensional input	Multiscale analysis	Large memory footprint
LR	Addressing classification and prediction task	Not suitable for non-linear relationships	May serve as benchmark model due to capture linear relationship	Limitations in learning spatio–temporal features
Boosting	High prediction accuracy	Lack of interpretability	Scalability	High computing requirements
ARIMA	Able to handle stationary and non-stationary time series data	Historical data requirements	Forecast future floods trends based on time series analysis	Unable to capture long-term trends
ANFIS	Manage model uncertainty	Lack of interpretability	High computational efficiency	Large number of parameters makes it error prone
KNN	Versatile for single and multi-objective problems	Difficulty to capture complex patterns and features	Offers efficient solutions for regression or classification tasks	Performance suffers with high-dimensional data
LSTM	Long-term data pattern identification	Mathematical complexity	Hybridization	Requires a large training dataset
RF	High prediction accuracy	Instability	Able to handle missing data	Prone to overfitting
SVM+SVR	High prediction accuracy	Unable to deal with noisy data	Hybridization	Difficultly in learning large dataset
MLP	Able to handle nonlinear complex data and noisy environments	Unable to handle complex relationships	Uncertainty analysis in probabilistic forecasting	Prone to overfitting
*ANN	Able to handle noisy environments	Unable to handle dynamic and complex rainfall–runoff relationships	Potential for use in transfer learning due to pattern recognition abilities	Lack of interpretability
Hybrid methods	Accurate and reliable prediction	Requires specialized expertise	Flexible modelling	Complexity in the system

characterized by highly complex mathematical expressions. These expressions include high computational needs, difficulty in tuning, and issues with interpreting the models.

Regarding opportunity, hybridization of different ML models such as LSTM and SVM could bring an increase in accuracy, speed, and reliability for short-term FF. The ANFIS model offers high computational efficiency, which presents an opportunity for forecasting. LR is used as a benchmark model for comparison with other models due to its ability to capture linear relationships and its ease of interpretation (Chau et al. 2005; Wu and Chau 2006a,b), while ARIMA can leverage historical data and forecast future flood trends based on time series analysis. Other models like KNN can efficiently perform regression or classification tasks. In general, hybrid approaches have benefits such as model flexibility (in term of linking with other models), an ability to capture complex patterns, and an ability to include concurrent processes. CNN and Boosting have strength in handling multi-scale datasets (Panahi et al. 2020). RF becomes useful if there are missing data in a simulation, and MLP can provide an uncertainty analysis for probabilistic forecasting. ANN is promis-

ing for transfer learning (a pre-trained model on one task is fine-tuned on new related task) because of its ability to accurately recognize patterns.

In terms of threats, all ML models were found to have deficiencies. These included: large memory usage, issues learning spatio-temporal features and long-term trends, high computing requirements, difficulties with learning, a lack of interpretability, and a tendency for overfitting.

When combining the multi-criteria and SWOT analysis, the hybrid, RF, and ANFIS approaches showed excellent performance. In addition, SVM, ANN, MLP, LSTM, and CNN achieved good performance, while Boosting displayed average performance. Other methods like LR, ARIMA, and KNN exhibited weak performance. Although this analysis provides a qualitative performance evaluation, uncertainties relate to similarity among ML methods and difficulty in decision-making (Hamitouche and Molina 2022; Petousi et al. 2017). In addition, this analysis lacks objectivity, as it can be influenced by individual perceptions and opinions. To address these factors, a comparative performance analysis provides a quantitative performance evaluation that strengthens the SWOT analysis, allowing for selection of the most suitable ML method for short-term FF across various lead times.

The comparative analysis of stand-alone and hybrid ML approaches is based on statistical parameters such as NSE, R^2, and NRMSE. The findings of this analysis are illustrated in Figs. 4 and 5. For 1–6 h lead times, most of the ML approaches such as CNN, ANFIS, MLP, KNN, hybrid, LSTM, ANN, and SVM have an excellent performance as shown in Fig. 4(a) (see, e.g., NSE and R^2 values greater than 0.85 and NRMSE values less than 15%). ARIMA, Boosting, and RF give a good performance with values of NSE and R^2 within 0.71–0.85 and NRMSE within 16–30%. LR, however, has an average performance with NSE and R^2 within 0.65–0.69 and NRMSE of 49%. This suggests that most of the ML models are suitable for making short-term predictions but their performance decreases as the lead time increases (Venkatesan and Mahindrakar 2019; Yadav et al. 2016; Zhou et al. 2023).

For 6–12 h lead times, the performance of hybrid, ANFIS, ANN, and LSTM remains solid as shown in Fig. 4(b) and captured by NSE and R^2 values greater than 0.85 and NRMSE below 15%. However, these values decrease slightly as lead times increase. MLP, SVM, RF, and KNN exhibit a good performance with NSE and R^2 spanning 0.71–0.85 and NRMSE within the range of 16–30%. LR gave an average performance with values of 0.61 for NSE, 0.65 for R^2, and 44% for NRMSE. CNN, Boosting, and ARIMA have been excluded since no examples were found in the literature for this time scale.

Figure 5(a) illustrates the results for lead times of 12–24 h. Here, SVM, hybrid, and RF give the best performance, with NSE and R^2 values greater than 0.85 and NRMSE values below 15%. Other algorithms such as ANFIS, LSTM, ANN, and MLP demonstrate a good performance, with NSE and R^2 values within 0.71–0.85 and NRMSE falling within the range of 16–30%. In addition, KNN gives an average performance, achieving values of NSE and R^2 within the range of 0.51–0.7 and NRMSE spanning 31–50%, respectively. CNN, LR, and ARIMA have been excluded since no examples were found in the literature for this time scale.

For 24–48 h lead times, the hybrid and RF approaches demonstrate an excellent performance, as indicated by NSE and R^2 values exceeding 0.85 and NRMSE values below 15% [see Fig. 5(b)]. These approaches gave a consistently greater accuracy and performance at all lead times than the other stand-alone models (Arabameri et al. 2020; Ebtehaj and Bonakdari 2022). Furthermore, ANN and ANFIS gave a good performance in forecasting with

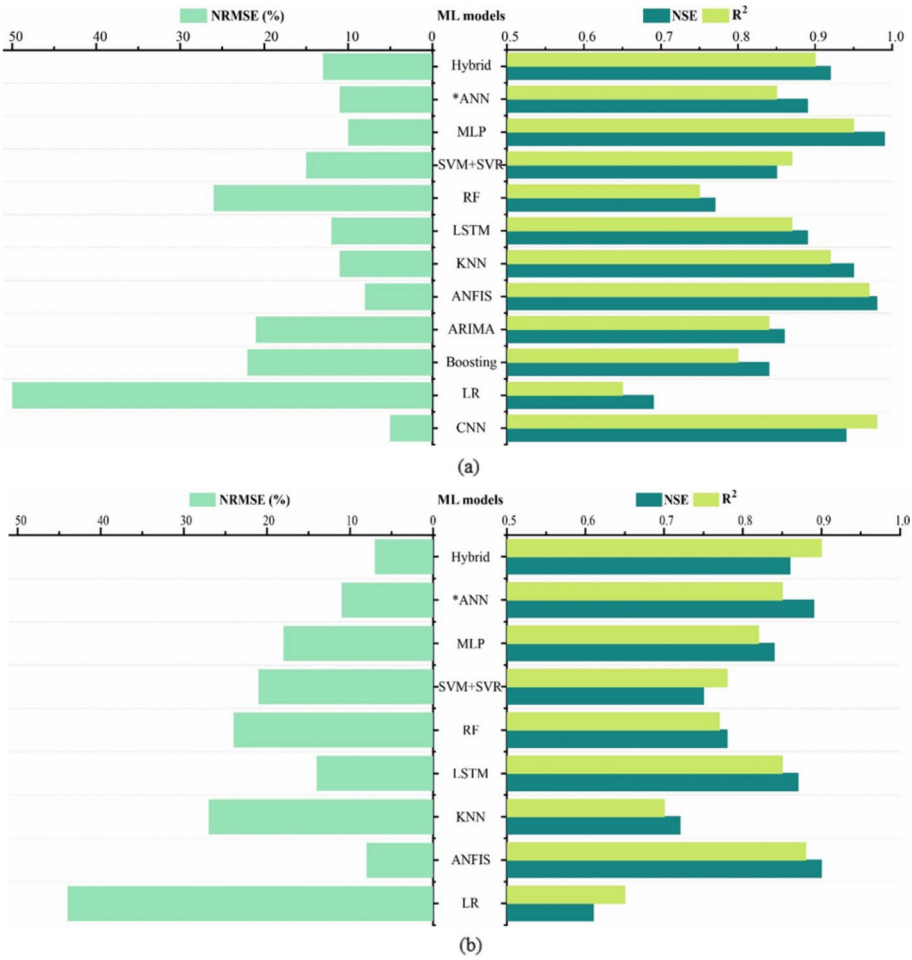


Fig. 4 Comparative assessment of ML approaches for short-term FF with lead times of (**a**) 1–6 h (**b**) 6–12 h

NSE and R^2 within range of 0.71–0.85 and NRMSE within 16–30%. Meanwhile, ARIMA and LSTM gave an average performance for this time scale with NSE and R^2 spanning 0.51–0.7 and NRMSE of 31–50%. LR, KNN, and MLP gave comparatively weaker performances as indicated by their NSE and R^2 values below 0.51 and NRMSE exceeding 50%. For CNN, Boosting, and SVM, no performance parameters have been found for this specific time frame in the literature.

A comprehensive performance evaluation of ML approaches across these lead times and based on the aforementioned analyses is shown in Fig. 6. Performance categories are: excellent, good, average, and weak. For 1–6 h lead time, most of the ML approaches (CNN, ANFIS, MLP, KNN, hybrid, LSTM, ANN, and SVM) give an excellent performance. ARIMA, Boosting, and RF, meanwhile, give a good performance. The strong performances can be attributed to the ability of models to capture data patterns on a short time scale for model training, resulting in accurate forecasting. The most promising ML models for this

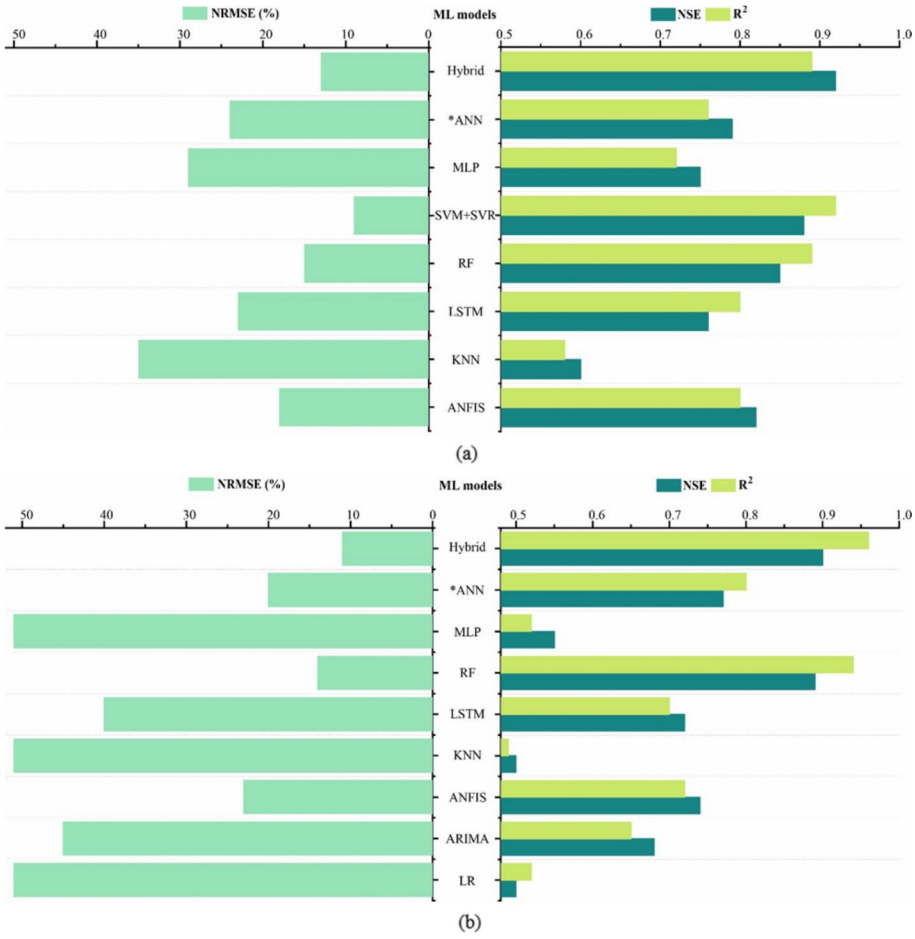


Fig. 5 Comparative assessment of ML approaches for short-term FF with lead times of (**a**) 12–24 h (**b**) 24–48 h

specific time scale include CNN, ANFIS, and MLP. Among these, CNN has been found to be a powerful tool for fast simulation of urban pluvial floods and has contributed to early warning systems (Liao et al. 2023). ANFIS appears to be more accurate and reliable than MLP for short-term FF (Rezaeianzadeh et al. 2014). Moreover, ANFIS has a lower complexity and requires less computation time compared to MLP. Among those methods with a good performance, ARIMA is superior due to its higher transparency and manageability.

For lead times of 6–12 h, the most promising ML approaches include hybrid, ANN, ANFIS, and LSTM. Meanwhile, MLP, SVM, and KNN give a good performance as shown in Fig. 6. The hybrid, ANFIS, and LSTM approaches, in particular, benefit from an ability to handle nonlinear relationships between input and output variables, which can enhance the accuracy of the model forecast. Furthermore, the hybrid approach outperforms LSTM and ANFIS in terms of transparency and manageability, as illustrated in Fig. 3. ANFIS compares favorably to LSTM in requiring less computation time and having a lower complexity. As

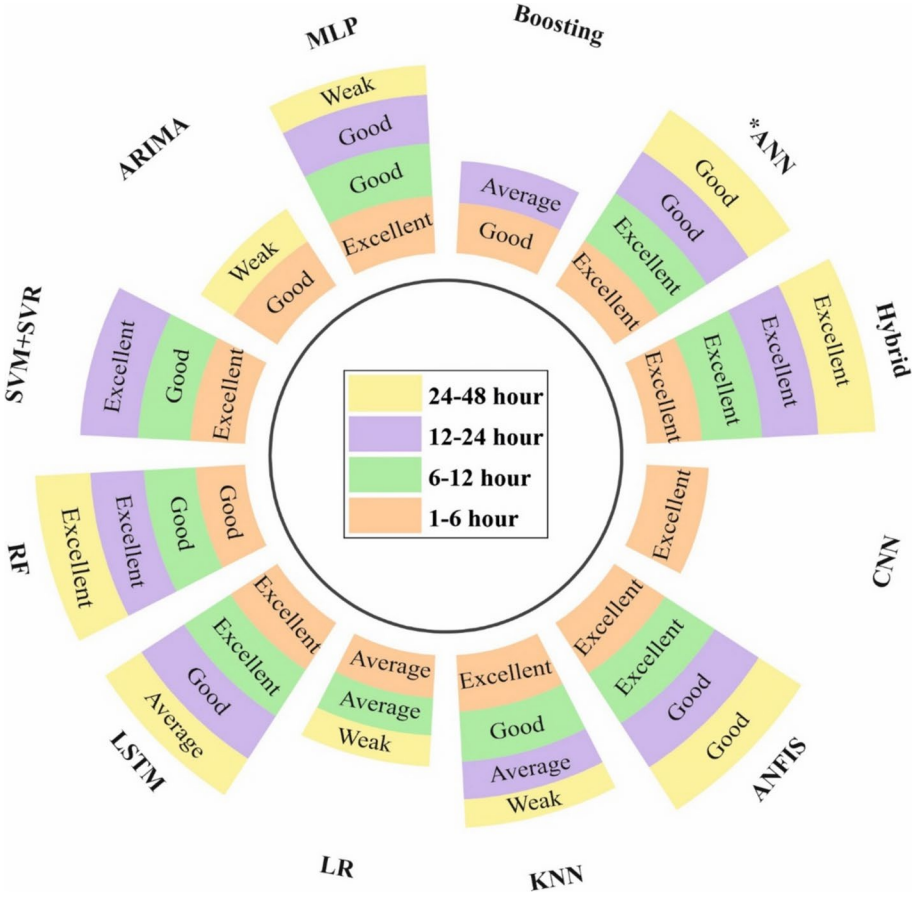


Fig. 6 Performance evaluations of ML approaches at different lead times

a result of its good performance, ANFIS is highly recommended for short-term predictions (Dazzi et al. 2021). LSTM has been shown to be accurate and reliable in the prediction of peak values. However, this requires a longer training time compared to other models (Le et al. 2019). MLP, SVM, and KNN exhibited a good performance with MLP outperforming SVM and KNN in terms of reliability and reduced computation time. The performance of MLP, however, appears to decrease as lead time increases (Macharyulu et al. 2022). Comparing SVM to KNN in forecasting for this specific time scale, SVM has exhibited a higher accuracy [see Fig. 4(b)].

For the 12–24 h lead times, the hybrid, SVM, and RF approaches give an excellent performance, while ANFIS, LSTM, and ANN have a good performance (Fig. 6). Specifically, the hybrid, SVM, and RF approaches perform well in handling large and complex datasets during model training and testing, particularly at longer lead times. Among these, the hybrid approach outperforms the others in demonstrating higher accuracy and shorter computation time during modeling (Fig. 3). Furthermore, hybrid approaches typically have an edge in enhancing transparency through the combination of suitable models. SVM and RF have shown a remarkable performance in forecasting, while SVM outperforms RF in accuracy.

Other models such as ANFIS, LSTM, and ANN have had a good performance due to their generalization abilities. Among these, LSTM stands out in its handling of complex data and the ability to deliver reliable outcomes at up to 24 h lead times. In general, all models performed admirably. However, ANFIS emerged as the most effective within this time frame. This can be attributed to a low level of complexity and computation time, as well as a higher reliability. ANFIS is a based on an artificial neural network, where a fuzzy inference system (FIS) is combined with an ANN to improve speed, which showed effective outcomes for this time scale (Rezaeianzadeh et al. 2014; Venkatesan and Mahindrakar 2020).

For the 24–48 h lead times, hybrid approaches and RF gave excellent performances, while ANN and ANFIS had good performances (Fig. 6). For these longer lead times, the advantages of hybrid methods in forecasting are apparent. By combining two or more stand-alone ML models, hybrid approaches can incorporate many input parameters, processes, and characteristics, thereby capturing the complexity of datasets (e.g., hydrological, meteorological, and topographical aspects) and providing highly accurate predictions [see Fig. 5(b)]. Hybrid approaches that integrate physical, stochastic, and ML models were shown to overcome the drawbacks of ML such as handling nonlinearity, capturing spatio-temporal dynamics, and data gaps. This enhanced the prediction accuracy as well as lead time. Another advantage shown by some hybrid models is the cost-effective computation for all lead times. RF also stands out for its ability to generalize and to handle high-dimensional data, providing more accurate predictions at higher lead times.

Among the moderately performing ML models, ANN gave a good performance, showcasing SWOT characteristics such as higher transparency and manageability as well as low computation time, in comparison to ANFIS (Fig. 3). When looking at other ML models, ANFIS provided accurate predictions at 24–48 h lead times. Both algorithms, however, experienced a decrease in performance as the lead time increased, indicating a limited ability to capture the complex relationship between input and output variables at higher lead times.

Looking at the overall performance derived from SWOT as well as the comparable analysis of several ML models with different lead times, hybrid methods outperformed their stand-alone counterparts. Hybridization may, thus, increase the accuracy and reliability of short-term FF across all lead times considered. Some stand-alone ML models have been effective in hybridization and improved the forecasting capability across lead times (see Table S2 in the Supplementary Material). Moreover, Table S2 (in the Supplementary Material) also provides the reasons for the hybridization success of each ML model. These models, including ANN, SVM, RF, LSTM, KNN, and GA, can be combined with other ML models to develop hybrid approaches for more accurate prediction.

4 Discussion

This study explores ML approaches for short-term FF at varying lead times. Using two methods of analyses, we evaluate various models, including CNN, LR, Boosting, ARIMA, ANFIS, KNN, LSTM, RF, SVM, MLP, ANN, as well as hybrid approaches. Each approach has different performance characteristics in each lead time span. Overall, this study reveals that some ML approaches give a better performance compared to others across all lead times, whereas the performance of other ML approaches strongly depends on lead time.

Specifically, hybrid methods and ANN, ANFIS, LSTM, RF, and SVM achieve a higher performance compared to other approaches across all lead times. Most approaches, however, show a great potential for forecasting, especially at short lead times.

Lead times are crucial in short-term FF as they allow for effective and timely response measures to mitigate flood risks and spare communities from damage. Consistent with other studies (Kurian et al. 2020; Saleh et al. 2016; Yang et al. 2015), we found that the model forecast performance decreases as lead time increases. At short lead times, uncertainties and noise in input datasets (such as precipitation and runoff) likely manifest themselves with smaller errors in the results. In contrast, forecasts with higher lead times may show poor performance due to higher errors and uncertainties in highly variable parameters (Roodsari et al. 2019). The NSE and R^2 values of models decrease and NRSME increases with lead times due to potential data limitations and challenges in capturing the complex dynamics over large periods (Mosavi et al. 2018).

It is important to consider that model performance not only varies with changes in lead times, but also depends on other factors such as (quality of) input parameters and the basin. In an ideal model comparison, factors such as the input datasets and study sites would be the same. This should be considered when interpreting the results of this study.

The selection and quality of input parameters are very important for the performance of ML in short-term FF, impacting their ability to capture key hydrological dynamics and to predict flood events. These parameters play a very crucial role in hydrological studies (Maspo et al. 2020; Singh et al. 2022). In Roodsari et al. (2019), inclusion of meteorological and hydrological input parameters increased the predictability performance of the ML model. In another study, Badrzadeh et al. (2015) showed that the addition of more effective input parameters significantly increased the accuracy of the model and also decreased the model complexity and number of input parameters. Overall, the choice of input parameters including environmental, socio-economical, and meteorological related variables impacts the reliability of predictions, which is essential for effective flood risk management.

The performance of ML also depends on the study site. This implies that underlying geographical conditions impact the effectiveness of the model. Each basin has different characteristics such as topography, land use cover, data availability, and climatic factors that affect the flooding process and also the model performance (Moghim et al. 2023). Again, the quality and quantity of data plays an important role. While for some basins extensive data exist, other basins have limited data. Therefore, Roodsari et al. (2019) suggest that model selection is very important in short-term FF and should align with characteristics of the basin as well as the available data (Kumar et al. 2023). Assuming that most authors followed this suggestion, this study highlights several approaches for application of ML to short-term FF. The best results were found using a hybrid approach, especially for longer lead times (Kurian et al. 2020). To further improve the accuracy of short-term FF using ML, hybridization (by integrating two or more ML methods or combining ML with a physics-based model) is a way forward. The hybridization of ML with a physics-based model remains less popular due to high computation costs and high model complexity compared to the hybridization of two or more ML methods (Mosavi et al. 2018).

5 Conclusion

To mitigate the impacts of floods, structural (e.g., dikes, reservoirs) and non-structural (early warning systems) measures can bring benefits. Short-term FF offers timely information that can be integrated into such early warning systems. However, the selection of ML models for short-term FF can be complicated by the impact of lead times on performance. This study evaluates the performance of selected ML models for short-term FF at various lead times.

First, a SWOT analysis was employed to evaluate the suitability of ML models for short-term FF based on strengths, weaknesses, opportunities, and threats as well as six additional criteria. This revealed that hybrid methods as well as RF and ANFIS gave an excellent performance; while SVM, LSTM, ANN, MLP, and CNN exhibited a good performance in short-term FF. Finally, Boosting had an average performance while LR, ARIMA, and KNN showed weaker results.

Second, a comparative performance analysis of stand-alone and hybrid ML approaches was performed using statistical parameters such as NSE, R^2, and NRMSE for short-term FF at different lead times. For lead time of 1–6 h, many ML approaches (e.g., CNN, ANFIS, MLP, KNN, hybrid, LSTM, and SVM) gave an excellent forecasting performance with NSE and R^2 values exceeding 0.85 and NRMSE below 15%. In addition, ARIMA, Boosting, and RF gave a good performance with NSE and R^2 within the range of 0.71–0.85 and NRMSE within 16–30%. For 6–12 h lead times, hybrid, ANFIS, ANN, and LSTM gave an excellent performance as captured by NSE and R^2 values above 0.85 and NRMSE below 15%. Other algorithms such as MLP, SVM, RF, and KNN gave a good performance with NSE and R^2 within 0.71–0.85 and an NRMSE of 16–30%. At 12–24 h lead times, SVM, hybrid, and RF performed excellently, as indicated by NSE and R^2 values exceeding 0.85 and NRMSE below 15%. Moreover, other models such as ANFIS, LSTM, ANN, and MLP exhibited good results, with NSE and R^2 values within the range of 0.71–0.85 and NRMSE of 16–30%. At 24–48 h lead times, hybrid models as well as RF demonstrated excellent forecasting performance with NSE and R^2 values exceeding 0.85 and NRMSE below 15%. ANN and ANFIS delivered a good performance achieving values of NSE and R^2 inside the range of 0.71–0.85 and NRMSE of 16–30%. Generally, the performance of most ML models decreased with increasing lead time.

Based on the two analyses, this study was able to identify models with a superior flood FF performance based on lead times. At 1–6 h lead times, CNN, ANFIS, and MLP performed excellently, with CNN leading. At 6–12 h, hybrid approaches, ANFIS, and LSTM excelled, with hybrid approaches outperforming the stand-alone models. At 12–24 h lead times, SVM, hybrid, and RF approaches had a superior performance, with SVM dominating the others. At lead times of 24–48 h, hybrid and RF approaches gave an excellent performance. However, again, hybrid approaches proved to have an edge. We believe that the consistently higher performance of hybrid approaches across all lead times is due to the inclusion of more input parameters, variables, and processes through the integration of two or more ML models. This study also found some effective stand-alone ML methods such as ANN, SVM, RF, GA, KNN, and LSTM, which provided better outcomes in hybridization with other ML models. Concluding, this manuscript can aid researchers, policy- and decision-makers in selecting appropriate ML models for short-term FF and early warning systems.

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Declarations

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Consent to Publish and Consent to Participate All authors have read and agreed to the published version of the manuscript.

Competing Interests The authors declare no financial of interest.

Conflicts of Interest The authors declare no conflict of interest.

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